



IoT and IoT in Medicine and Healthcare sector

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ABSTRACT

This article constitutes a review of the state of the art of the Internet of Things (IoT) and its use in the field of Medicine and healthcare in general. A definition of IoT is presented, the main characteristics of said technology are defined and its main application areas are explored. Among these is healthcare and medicine. IoT devices that are of great interest in this field are also addressed, as well as what constitutes a smart health system and the benefits it represents for medical care.

1. Introduction

The Internet of Things (IoT) has emerged as a transformative force, reshaping the landscape of information technology and its applications. This article delves into the state of the art of IoT, exploring its definition, applications, and particularly focusing on its profound impact on healthcare. The reach of IoT extends across diverse industries, with substantial contributions to healthcare, smart cities, homes, industries, connected cars, agriculture, wearables, and more. Each sector harnesses the power of IoT to enhance efficiency, productivity, and user experience. The impact of IoT in healthcare is particularly noteworthy, ushering in a new era of patient-centric, data-driven, and remotely monitored healthcare services. From remote patient monitoring to the management of hospital operations, IoT devices play a pivotal role. Smart healthcare systems, integrated with artificial intelligence (AI) and cloud computing, bring about a paradigm shift in disease diagnosis, treatment, and overall healthcare management. The convergence of IoT and medical applications, often referred to as the Internet of Medical Things (IoMT), represents a fundamental development. IoMT facilitates remote patient monitoring, ingestible sensors, robotic-assisted surgeries, and other innovations, contributing significantly to the improvement of patient care and healthcare delivery. While the promises of IoT in healthcare are immense, challenges exist. Ensuring data accuracy, addressing privacy concerns, fostering user-friendly interfaces, and navigating ethical considerations are crucial aspects that demand attention. Collaboration among stakeholders, adherence to security protocols, and a commitment to ethical data practices will be imperative for realizing the full potential of IoT in healthcare.

2. IoT Definition

According to the European Internet of Things Research Cluster (IERC), The Internet of Things (IoT) is defined as a global infrastructure for the information society, which allows the execution of advanced



services through the physical and/or virtual interconnection of things through interoperable information and communication technologies that currently exist, but that evolve over time.

The term thing in IoT refers to any element that can be connected to the Internet, whether it is an appliance, a piece of furniture, a sensor or a network of sensors to monitor the temperature and humidity of a plantation in a greenhouse, an electricity meter in a home, a video surveillance camera in an airport, etc. A software robot (bot), or simply a person through the devices that they can carry, such as mobile phones or wearable devices, such as bracelets or smart clothing, etc.(Víctor Manuel Brea Sánchez, 2018)

The Internet of Things (IoT) is a network of individually identifiable connected things that offers services for intelligent computing. IoT objects are also referred to as “smart things,” which make it possible to carry out routine tasks in a logical manner. The Internet of Things also benefits human communication in good ways. IoT’s basic tenet is the provision of seamless, all-pervasive connectivity between objects and people. In a nutshell, IoT is a system in which things are linked in such a way that they may intelligently interact with one another and with people.(Fuior et al., 2023)

The IoT enables physical objects to see, hear, think and perform jobs by having them “talk” together, to share information and to coordinate decisions. The IoT transforms these objects from being traditional to smart by exploiting its underlying technologies such as ubiquitous and pervasive computing, embedded devices, communication technologies, sensor networks, Internet protocols and applications. Smart objects along with their supposed tasks constitute domain specific applications (vertical markets) while ubiquitous computing and analytical services form application domain independent services (horizontal markets). Fig. 1 illustrates the overall concept of the IoT in which every domain specific application is interacting with domain independent services, whereas in each domain sensors and actuators communicate directly with each other.(Al-Fuqaha et al., 2015)



Fig. 1 The overall picture of IoT emphasizing the vertical markets and the horizontal integration between them.

3. Characteristics of The IoT

The IoT represents a promising future technology that shows some features as follows:

- **Large scale:** IoT devices are increased in billions. These devices need to be managed to be able to communicate with each other. The management of data generated from these devices and their interpretation for application purposes becomes critical.
- **Intelligence:** With the integration of software algorithms and hardware, IoT devices become smart. These intelligence capabilities make IoT devices interact in an intelligent way in certain situations. Although the popularity of smart technologies, intelligence in the IoT is only means of interaction between devices, while user and device interactions are achieved by usual input methods and graphical user interfaces.
- **Complex system:** IoT has billions of devices. Performing operations with this large scale of objects/things make the coordination process very complex due to constraints associated with memory, energy and time.
- **Dynamic environment:** IoT is a dynamic network in nature where objects are continually deployed; some new items joining the network while others leaving without determining network boundaries. The IoT devices can dynamically adapt to changing situations based on their operating conditions. For example, surveillance cameras can adapt their modes based on whether it is day or night. Cameras could switch from lower resolution to higher resolution modes when any motion is detected and alert nearby cameras to do the same.
- **Heterogeneity:** The IoT system involves different devices, platforms, operating systems, and services which are connected with each other using different protocols.
- **Limited energy:** Most IoT devices are small and lightweight with limited resources, so they are designed to work with minimal energy consumption.
- **Connectivity:** Connectivity enables network accessibility and compatibility. It empowers the IoT by bringing together everyday objects. It also provides new market opportunities for IoT that can be created by the networking of smart things and applications.
- **Context awareness:** There are many sensors that collect information about the physical and environmental features, therefore, sensor nodes get knowledge about the surrounding context, so it considered as context-aware. (Atlam et al., 2018)

4. IoT Architecture

The IoT should be capable of interconnecting billions or trillions of heterogeneous objects through the Internet, so there is a critical need for a flexible layered architecture. The ever-increasing number of proposed architectures has not yet converged to a reference model. Meanwhile, there are some projects like IoT-A which try to design a common architecture based on the analysis of the needs of researchers and the industry.

From the pool of proposed models, the basic model is a 3-layer architecture consisting of the Application, Network, and Perception Layers. Some other models have been proposed that add more abstraction to the IoT architecture. Fig. 2 illustrates some common architectures among them is the 5-layer model. Next, are provided a brief discussion on these five layers. (Al-Fuqaha et al., 2015)

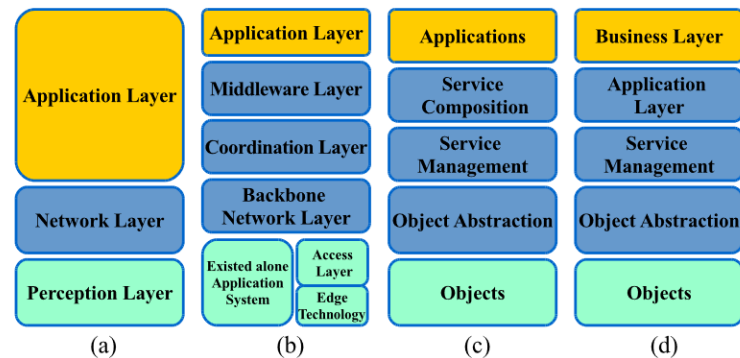


Fig. 2The IoT architecture. (a) Three-layer. (b) Middle-ware based. (c) SOA based. (d) Five-layer.

4.1. Objects Layer

The first layer, the Objects (devices) or perception layer, represents the physical sensors of the IoT that aim to collect and process information. This layer includes sensors and actuators to perform different functionalities such as querying location, temperature, weight, motion, vibration, acceleration, humidity, etc. Standardized plug-and-play mechanisms need to be used by the perception layer to configure heterogeneous objects. The perception layer digitizes and transfers data to the Object Abstraction layer through secure channels. The big data created by the IoT are initiated at this layer.

4.2. Object Abstraction Layer

Object Abstraction transfers data produced by the Objects layer to the Service Management layer through secure channels. Data can be transferred through various technologies such as RFID, 3G, GSM, UMTS, WiFi, Bluetooth Low Energy, infrared, ZigBee, etc. Furthermore, other functions like cloud computing and data management processes are handled at this layer.

4.3. Service Management Layer

Service Management or Middleware (pairing) layer pairs a service with its requester based on addresses and names. This layer enables the IoT application programmers to work with heterogeneous objects without consideration to a specific hardware platform. Also, this layer processes received data, makes decisions, and delivers the required services over the network wire protocols.

4.4. Application Layer

The application layer provides the services requested by customers. For instance, the application layer can provide temperature and air humidity measurements to the customer who asks for that data. The importance of this layer for the IoT is that it has the ability to provide high-quality smart services to meet

customer's needs. The application layer covers numerous vertical markets such as smart home, smart building, transportation, industrial automation and smart healthcare.

4.5. Business Layer

The business (management) layer manages the overall IoT system activities and services. The responsibilities of this layer are to build a business model, graphs, flowcharts, etc. based on the received data from the Application layer. It is also supposed to design, analyze, implement, evaluate, monitor, and develop IoT system related elements. The Business Layer makes it possible to support decision-making processes based on Big Data analysis. Moreover, this layer compares the output of each layer with the expected output to enhance services and maintain users' privacy.

5. Application areas of the internet of things

Applications based on the Internet of Things (IoT) are expanding quickly, which is causing the world to change. IoT's expansion has been a wonderful development in recent years. IoT is the interconnection of physical and digital items that have been fitted with sensors, software, and other technologies. IoT has impacted a wide range of industries, starting with the healthcare industry. It is now implantable, wearable, and portable, creating a pervasive and interactive world. It transforms the inanimate objects in our immediate environment into intelligent objects, resulting in the creation of an information environment that raises the standard of living for people.

IoT devices, for instance, monitor and gather vital measurements (such as blood pressure, blood sugar level, pulse, etc.) in real time, enabling emergency alerts to raise the patient's likelihood of survival. Furthermore, autonomous and self-driving cars assist drivers in getting where they are going by preventing them from going off the road or getting into accidents. Additionally, these definitions are expanded to include automatic emergency alerts for the closest roads as well as medical aid in the event of an accident. The Internet of Things also includes a wide range of contemporary industries, including the manufacturing, assembly, packaging, logistics, smart city, and aviation sectors. Figure 3 depicts a few of the most important IoT-based application sectors in the fields of health, business, communication, and entertainment. (Fuior et al., 2023)

5.1. Healthcare

The IoT has brought many benefits and opportunities to the field of healthcare. It helps to develop and improve healthcare services and keep the field innovative. For instance, intelligent drug/medicine control and hospital management. In addition, the IoT add more benefits by monitoring the individual health in real-time. Also, ambulances can be immediately dispatched to accident scenes and patients can be monitored at their homes just as effectively as in hospitals. For example, a doctor can immediately be informed if the patient suffers a heart attack.(Mangla et al., 2021)



Fig. 3 Important application areas of the internet of things.

5.2. Smart City

The concept of the smart city is used to describe the better use of public resources, increasing the quality of service presented to the citizens, and at the same time reducing operational costs of public administrations. The IoT provides several benefits in the management and optimization of public services, such as transport and parking, lighting, surveillance and maintenance of public areas, preservation of cultural heritage, and garbage collection. Furthermore, the availability of different types of data collected by IoT devices can be used to enhance the awareness of people about the status of their city and stimulate the active participation of the citizens in the management of public administration. (Mangla et al., 2021)

5.3. Smart Home

Currently, people can install smart appliances inside their homes to control many of the house tasks. These intelligent devices have the option of remote control, which eliminates the need of being near the device. Therefore, these devices have enabled the automation of home activities by the adoption of various embedded devices. (Mangla et al., 2021)

5.4. Connected Industry

The connected industry is the vision of a manufacturing environment where every machine can communicate with all other machines across the plant. The connected industry with IoT will connect, monitor and control virtually anything, anywhere to provide operational productivity and profitability. In addition, the integration of IoT with sensor networks, wireless connectivity, innovative hardware and machine-to-

machine communication will completely transform the conventional automation process of industries. (Mangla et al., 2021)

5.5. Connected Car

Connected cars are equipped with Internet access and can share their access with others, just like connecting to a wireless network in a home or office. More vehicles are starting to come equipped with this functionality, so be prepared to see more apps included in future cars. The connected car is considered as the best way to minimize accidents such that a pilot can operate the car remotely to minimize car accidents and reduce human errors. These driverless cars can provide functions more than just safety such as they can save valuable time, reduce the stress of driving etc. Some studies reveal that by 2040, driverless cars will account for up to 75 percent of cars on the road worldwide. (Mangla et al., 2021)

5.6. Smart Parking

In recent time, smart parking sensors are attached in parking space to detect the arrival and departure of vehicles. It provides an efficient management solution to help drivers to save time and fuel. It provides the drivers with the accurate information about parking spaces and keeps the traffic system smooth. It also enables the facility of deployment to book parking space directly from the vehicle. (Mangla et al., 2021)

5.7. Smart Energy and Smart Grid

The IoT provides more information about the behaviors of electricity suppliers and consumers in an automated way to improve the energy efficiency. It also provides consumers with smart management of energy consumption such as smart meters, smart appliances, renewable energy resources. (Mangla et al., 2021)

5.8. Environmental Monitoring

The key element of the IoT system is sensors which collect information about the surrounding environment. Therefore, with the IoT, a high-speed information system can be provided. This allows the entity that monitors wide-area environments and sensors deployed in the area to convey a huge amount of data easily such as pollution source monitoring, water quality monitoring, air quality monitoring. (Mangla et al., 2021)

5.9. Smart Agriculture

With the presence of sensors in everywhere, farmers can use the huge collected information to yield a better return on investment. Sensing for soil moisture and nutrients, controlling water usage for plant growth and determining custom fertilizer are some simple uses of IoT in the agriculture. In addition, many wireless technologies were used in the agriculture such as remote sensing, global positioning system and geographical information system. This in turn will replace human labour with automatic machinery which will increase the productivity. (Mangla et al., 2021)

5.10. Wearables

Watches are no longer just for telling time. Smartwatches have turned our wrists into a smartphone by enabling text messaging, phone calls, and more. Many other devices are used to give us more information about our workouts such as Fitbit, Jawbone and others. (Mangla et al., 2021)

6. IoT in Medicine: Medical IoT

The most important IoT field is medical and health care, which is poised to alter the healthcare industry over the next decade because of its vast potential for various applications, ranging from hospital-centric to patient-centric. Patients are getting more involved in their treatment due to the IoT technology, which would allow them to reach their doctors, organize appointments, and access their health records via a portal. The health workers and individuals could also monitor their health from home in real-time using the home monitoring systems, providing benefits for elderly people and patients with long-term diseases (Verma et al., 2022)

In the medical field, IoT can be useful in remote patient monitoring (monitoring blood pressure, checking heart rate, checking biometric parameters, or even checking hearing aids), it can be used in the management of diseases in chronic patients or in case of medical emergencies. The advantages of IoT systems used in medicine are that they can continuously and reliably monitor patients and facilitate the digital storage of patient's personal health information. This type of technology helps in the formation of medical databases and their interconnection for a much better management of patient care. Studies show that at this moment, IoT technology will lead to the greatest advances in medicine and will generate revolutionary treatments for patients.

IoMT or the Internet of Medical Objects represents the Internet of Things used in healthcare applications. There are several IoMT solutions on the market but also at the study level that include sensors, wearable systems, including remote access to medical services and monitoring systems of daily activities or hospital systems that increase the quality of patient care, thus fully covering the need for patient care (Fuor et al., 2023)

7. Applications of IoT devices in healthcare (Mangla et al., 2021)

Various applications of the IoT in healthcare system which can give a new dimension to the patient treatment and healthcare services are discussed below.

7.1. Monitoring of Patients Remotely

Patient monitoring from remote locations is possible with the help of IoT devices. This reduces the number of cases of hospital admissions which, in turn, reduces the expenses. After a major illness or surgery, when the patients are discharged from the hospitals, doctors can keep track of the health of the patients and reduce re-admissions. In rural areas where the healthcare facilities are poor,

doctors can monitor the patients sitting at a remote location and can reduce the death rates. The expenses incurred for traveling long distance for consultations, hospitalizations, etc., can be reduced. Even IoT devices can alert the healthcare professionals if the patients are not taking medicines regularly or taking wrong medications. With the continuous monitoring of the patients, health risks can be reduced drastically.

7.2. Management of Hospital Operations

Hospital operations involves management of various equipment and drugs. It is difficult to manage this equipment in an optimal way as the doctors are always busy with their patients. By means of IoT devices, management of equipment becomes much easier. As the equipment are regularly monitored, fault in that equipment can be easily detected and rectified. Also, IoT devices can alert the authorities about any outdated equipment which needs replacement. Most of the time, it is difficult to track how many available equipment are for use and the exact location of the equipment. IoT devices keep track of the available equipment and their locations which helps the healthcare professional to locate the equipment quickly which can save many lives. IoT devices also alert the hospital authorities about the cleanliness and hygiene of the surrounding environment which can lead to maintenance of a healthy environment which, in turn, prevent the spread of the diseases.

7.3. Monitoring of Glucose

With the advancement in technology several wearable connected devices and sensors are available in the market which can monitor glucose levels on real-time basis and update it on the connected devices. As the sugar levels are monitored on regular basis, any rise in sugar levels can be detected early and proper action can be taken at the right time before it affects the overall health.

7.4. Sensor Connected Inhaler

Usually, symptoms of an asthma attack appear few hours before the peak of air pollution. If the sensor connected inhalers are used by the patients, then it can detect triggering factors and alert the patients as well as the doctors about the possibility of an asthma attack through IoT connected devices, helping the doctors in instructing the patients when to stop inhalers.

7.5. Connected Contact

Diagnosis of some of the diseases can be made through by observing the changes that occur in the eyes. Contact lens is commonly used nowadays for eye sight problems and for enhancing the appearance of the eyes. With IoT-based contact lens, it can do wonders for humans. It not only aids in enhancement of eye sight it also monitors the changes that occurs in the eyes which can lead to early diagnosis of diseases. By monitoring the variations in the eye ball size, it can diagnose glaucoma at an early stage, any delay in detection of abnormalities may result in loss of eye sight.

7.6. Hearing Aid

Hearing aid has been used by many people around the world who have hearing issues and wants to enhance hearing. But due to the invention of IoT devices, hearing aids became smarter. It not only aids in hearing but also assist in other activities too. People even can hear their door bells ringing from a far place if it is connected by IoT devices. It can detect smoke if the smoke sensor is attached to it. These devices can filter out the noise effectively so people can enjoy music or personal conversations in a noisy environment without any disturbances. Also, several conversations can be listened simultaneously.

7.7. Coagulation of Blood

Clotting of blood plays a very important role in human body. To avoid serious medical conditions which may result in strokes, regular monitoring of blood clot level is essential. With IoT-enabled devices, it is possible to monitor the blood clot level on a regular basis. With the regular monitoring of clot levels, bleeding in the brain, strokes, heart attacks, etc., can be minimized by seeking medical attention at the right time. This can avoid unnecessary medications, hospital admissions, surgeries, etc., which, in turn, can save the medical expenses.

7.8. Depression Detection

With the advent of smart devices like wearable smart watches, it is easy to track depression levels. These smart devices monitor the depression levels and provides suggestions about the type of treatments available. Psychiatrists or psychologists can study about the mental conditions of the patients by observing their historical data of depression levels and can provide right treatments as the data is continuously monitored by the connected devices. With the invention of such smart devices, depression can be easily treated by studying the sleep patterns and depression levels of the patients.

7.9. Detection of Cancer

Early detection of cancer is very important to save the life of an individual. The most common and dangerous cancers seen among women is breast cancer, and if it is not detected at an early stage, it leads to fatal condition. With the advancement in new technologies, sensor-enabled bras are developed which makes use of Artificial Intelligence which, when worn, can detect abnormalities in breast tissues. These bras need not be worn throughout for detection; women can wear this bra for 2 to 12 hours per month. This wearable sensor embedded cloth does not produce any radiations or side effects. This can detect breast cancers at an early stage so that the treatment can start at an early stage for which complete cure is possible. These wearable devices reduce the cancer risk among women and can reduce the treatment expenses as the doctors and patients will be alerted about the abnormalities in the tissues.

7.10. Monitoring Parkinson

With IoT devices, regular monitoring of Parkinson's patients on real time is possible. Patients may suddenly face lack of movement which is very common in Parkinson disease called Freezing of Gait (FOG). This FOG stage can be detected by means of IoT devices and alerts can be given to doctors and patients. Even early detection of symptoms of this disease is also possible by the use of IoT devices. In future, such devices will help the Parkinson's patients to lead a better life without risking their lives.

7.11. Ingestible Sensors

Sensors are embedded into the pills which can be ingested by the patients. These sensors will monitor whether the patient has taken prescribed medicines at the right time. If the medicines are not taken properly, it will be alerting through the connected devices. Most of the times patients does not follow doctor's prescription, they either miss to take medicines on time or they may take overdose of the medicines. They may not follow the proper time gap for each medicine. All these issues may lead to serious medical conditions and result in expensive medical treatments. These situations can be avoided by ingestible sensors which regularly monitors the patients and alerts them to take medicines on time. Taking care of the patients also becomes much easier with the support of such smart devices.

7.12. Surgery by Robotic Devices

Robots connected to IoT devices can perform complex surgeries with more precision and control by the supervision of doctors. Doctors even can instruct robots to perform surgeries sitting at a remote location. This can save many lives as quick action is possible even at a remote location. Many of the surgeries lasts for hours to complete which is a tiresome job for the doctors. In such situation, robots can perform the surgeries with much better precision. Complicated surgeries can be done successfully with the help of robotic devices connected to IoTs. This gives new dimensions to the classical surgery procedures.

7.13. Efficient Drug Management

Using IoT-enabled devices management of drugs can be done in an efficient manner starting from the production to the supply chain management. Radio Frequency Identification (RFID) tags are used for distribution purpose which also records a real-time record of the availability of the drugs and whether distribution is correctly done. The temperature during production process is also monitored to detect any variations. With the proper supply chain management, production cost can be reduced and patients will get good quality drugs at a reduced rate.



7.14. Body Scanning

A smart body scanner which scans the body and updates about the variations in body parameters can be used. The scanner is basically a full-length mirror with a 3D camera and a weigh scale which scans the entire body and stores the images and the data will be used for processing the results in a mobile application. This application will create a graphical chart about the variations in body dimensions and parameters which will be helpful in maintaining fitness.

7.15. Medical Waste Management

Hospital waste management is a difficult problem faced by the healthcare facilities throughout the world. If these bio medical wastes are not disposed properly. Sensor-enabled dustbins may be used which is connected through IoT devices. The sensors detect the amount of trash present and automated messages will be generated to dispose the waste when the garbage bin is full. Through the connected devices, autonomous robots can fetch the garbage and dispose it in a smart way. As robots are involved in disposing medical wastes, this may reduce the risk of spreading of the diseases.

7.16. Monitoring the Heart Rate

With the advancement in technology, regular monitoring of the heartbeat and pulse rate is possible by means of sensor-enabled IoT devices. Since it is a real-time monitoring, any variations in the heart rate or pulse rate will be alerted to the patients as well as doctors through connected devices. Immediate medical attention is possible which can save many lives.

7.17. Robot Nurse

Artificial Intelligence-enabled robots can assist the nurses in their day-to-day activities. The robots may be connected to IoT devices so that the activity of the robots can be scheduled and monitored by the nurses. Robots can assist the nurses in picking and placing the medicines and alerting the patients to take medicines on time. They can give personal care for the patients and also can clean the rooms and beds. Nurses can assign the tasks to robots and monitor it. These robots may be controlled by the patients and nurses by means of mobile devices. With these assistant robots, nurses can manage their workload in an efficient manner and provide better service to the patients.

8. IoT Healthcare wearable devices(Verma et al., 2022)

The current categories of IoT-based portable devices include body clothes (pants, underwear, and coats), heads (helmets and glasses), wrists (gloves, bracelets, and watches), somatosensory modulators, like, body and feet sensory control devices.

Further, with the help of data collected in the Kaggle repository, doctors may now use an AI enabled IoT in patients to detect a variety of illnesses, including diabetes, heart disease, and gait difficulties. The

Google Brain initiative's AI-powered diabetic eye disease detection is a current example of AI-based IoT based on Deep Learning, which employs neural networks to learn and execute a specific job through repetition and self-correction. Another example is next-generation wearables for blind people that utilize ultrasounds to identify impediments in the user's path and alert them so they may safely navigate items around them. Furthermore, AI wearables can aid fitness professionals with their everyday training activities by allowing them to track their progress. Bluetooth smart with. To summarise, artificial intelligence in wearables pushes the boundaries by supporting patients and clinicians with remote tracking, precautions, remote diagnostics, and guiding patients in making progressive decisions.

8.1. Wrist mounted wearables

The wrist-wearables are usually meant for the wrist to monitor physiology, having the benefits of miniaturization along with an increase in battery life for converting the raw signals to real-time interpretable information. Measuring arterial blood pressure appears to be a potential method for monitoring and controlling the prevalence of cardiovascular disease-affected patients. As a result, blood pressure monitoring has become one of the essential physiological measures in the ambulatory context for monitoring an individual's health condition. In this respect, wrist-based wearables can be a clinical tool with a high acceptance rate by the patient. Further, wrist-based devices are used to track daily activities such as patient monitoring, acceleration, rotation, gesture, and motion.

8.2. Head-mounted devices

The head-mounted devices are the visual tools having hands-free abilities, generally fixed to the person's head such as caps, glasses, and helmets which are recently applied in education, simulation, imaging, surgery as well as in navigation tools. Wearable systems consist of smart glasses that also fall under the category of the head-mounted computer having a display. Based on the literature, numerous smart eyeglasses are being developed for different applications like biosensors for sweat lactate utilizing bienzymatic Gel-Membrane through eyeglasses, EOG (electrooculography)-based human-wheelchair interface, health monitoring and medical use, computational eyeglasses for sensing drowsiness and fatigue, biosensing of tear for minerals and vitamin detection. Cavitas wearable sensors are fitted to body cavities like mouth guards and contact lenses that deliver data from the biological fluid. In addition, these mouth guard sensors have also been explored for real-time detection of substances in saliva. A mouth guard glucose sensor establishes on microelectrochemical system (MEMS) having enzyme membrane immobilized glucose oxidase was developed.

8.3. Smart clothing/e-textiles

Smart clothing, also known as e-textiles, is a growing interdisciplinary area of wearables having potent usage in fitness and healthcare. It comprises clothing material and conductive devices connected to sensing various environmental situations and responding to mechanical, chemical, or thermal fluctuations. The textile-based analytical system includes sensors utilized for biofluids analysis. A controlling unit, an actuator

and a sensor are used for designing smart textiles that are employed for monitoring human physical activity like pressure, body acceleration and motion, biomechanics and physiological signals.

An electronic shoe was designed to monitor walking ability like ground reaction forces, toe pressure, heel strike, and lateral plantar pressure, providing basic information for differentiating between the walk phases.

Additionally, a smart shirt-based biosensor was designed to measure the acceleration signals and electrocardiogram for real-time and regular health tracking. These wearable sensors have been developed to fit comfortably into a shirt with low power consumption and tiny size to decrease the battery size. Because of its mechanical flexibility and low cost, the biosensor using artificial biology technology offers the potential to monitor the environment and healthcare.

8.4. Chest mounted devices

The checking of postural and falling disability of patients is a top concern for health workers or caregivers. The two vigilant systems that are offered commercially for safety monitoring are the AlertOne medical alert system (“AlertOne: Medical Alert System” n.d.) as well as the Life Alert Classic via Life Alert Emergency Response Inc (“LIFE ALERT CLASSIC,” n.d.). A pendant is combined with a push button in these devices, and hitting the button sends the message wirelessly to a remote place. Similarly, the Wellcore system utilizes accelerator meter components and advanced microprocessors to monitor postural movement. This device helps in distinguishing between falls body and normal movements that further communicate to a remote center. Halo Monitoring TM invented the My Halo TM, a chest-worn device employed to monitor the temperature, sleep pattern and heart rate, etc. (“myHalo™ Personal Monitoring and Alert System,” n.d.; Patel et al., 2012). In summary, a device with an integrated system on a smartphone and a balance sensor that prompts automated dialling on an emergency number in the fall incident would be extremely valuable. (Verma et al., 2022)

9. IoT Smart Healthcare System

The IoT is really a ground-breaking paradigm of communication which focuses on building the invisible and inventive structure for linking wealth to electronic devices through the internet. Exchanging data from tiny home appliances to huge industrial devices, the idea behind IoT has reduced the breach between the physical and the digital world. In other words, IoT which is a network of interconnected devices that sense, accumulate, and transfer data over the internet can interactively perform the task with the help of real-world entities. There is a variety of data available on the internet which can further be used as an offline data source for training different deep learning or machine learning applications. But to turn the city into a smart city and to automate various other applications like smart healthcare, there is a need to capture the real-time data, gathered that data and process it and take the immediate required action or reduce the loss and this is what “Internet of Things”. This has given birth to many other terms like IOHT, IOCT, and so on as its major application. It is therefore designed to allow the internet with extra accessible and pervasive. Sensors in IoT devices have a vast significance which inevitably increases the direct integration between both the wearable devices as well as the physical



environment. The potential of sensors which determines the changes in its surrounding and connecting the digital world with the physical world more efficiently by taking autonomous decisions, which further extends its scope in other areas of use like healthcare monitoring systems, household gadgets, large industrialized devices, and smart cities, IoT systems are available in several necessary application areas. (Mangla et al., 2021)

10. Applications of Smart Healthcare System (Mangla et al., 2021)

In this world of digitization, almost every sector can be easily visualized through wireless technologies. With the advancements in technologies as the IoT, AI, and cloud computing, the smart healthcare system has been improvised to a new level from each perspective. The healthcare sector contributes a major share of revenue and also generates huge employment leads to the development of a country.

Smart healthcare along with AI and cloud computing elevated the functioning of the overall healthcare sector. Nowadays, a large number of computerized devices and apps based on IoT deep learning and AI are available at finger touch. With the increasing population, the need for advanced health facilities became a challenge and extensive research seemed going to develop a better healthcare system. Advanced technologies inclusive of Information and Communication Technology (ICT) assisted the ease of access, transfer, interpret, and gather the same from different signals—both permanent and remote. IoT built up an intelligent system along with the other technologies inclusive of sensor technologies, wireless communications, nanoelectronics, and others. Another technology of nanomaterials boosted up the healthcare system by removing the cons of conventional materials and improvising the overall system functioning. These potential devices allowed flexible communication between human beings or coordination between machines and interconnection between the patient and the machine.

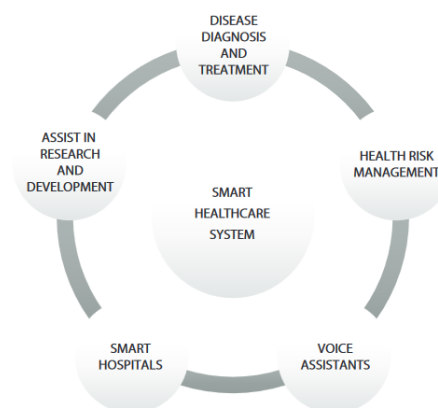


Fig. 4 Smart healthcare system with its dimensions.

10.1. Disease Diagnosis and Treatment

A smart healthcare system rooted its branches in the diagnosis of a disease and appropriate treatment as per the necessity. Cloud computing and AI improvised pathological issues. Ease of online appointments for testing in labs and getting the reports through the internet reduced the stress of waiting long hours. In addition, several apps and sites are available inclusive of Diagnosis and Lab Values Pro that assist the patients to understand the reports and the standard normal values of a particular examination. EEG diagnosis tools have been successfully used in the detection of various disorders inclusive of epilepsy, stroke, Alzheimer's, and brain injuries. A new program field-programmable gate array (FPGA) analysis system along with mobile application has been reported that updates the pathological data of a patient to its wearable device to monitor the abrupt changes further and finally controlling the fatal impact of a disease. Further, another system has been developed to sense the data of the patient and uploaded the data once it hits the threshold values which further transferred to an analysis system named FPGA system. The data has been analyzed through FPGA system and can be viewed on wearable IoT device of the patient.

10.2. Health Risk Monitoring

Big data analysis and deep learning made possible to monitor health issues and manage the risk developed for a particular disease. Various gadgets and wearables have been reported that helps in monitoring health. For instance, smartwatches, smart bracelets, smart headsets, smart clothing, and wearable sensor patches serve as a platform for monitoring temperature, heart rate, blood pressure, and others. In addition, pillboxes are found to be another accomplice that reminds the patient of their daily dosages. Adhere to Tech's smart pillbox, iMedipac, MedMinder, and smart pillboxes are some of the examples of pillboxes. Not confined to pill bottles but ingestible biosensors have also been reported that monitors drinking of water and pill pickups ultimately promotes medication adherence. RFID and computer vision technologies also assist dosage patterns. Voice disorder became prevalent among lawyers, teachers and singer's these days which can be now be monitored through computed linear prediction (LP) analysis that interprets the irregularities in vocal fold vibrations. An IoT-based ECG wearable node along three electrodes have been proposed eliminating the need for mobile application [4].

10.3. Voice Assistants

In this tech-savvy world, voice technology has defeated the stress and loneliness of elderly and loner patients. Alexa, Apple Siri found to act like virtual assistants that not only remind the patients for their dosage but also monitors the health. Chatbots were found to be another emerging field that has been designed to communicate with people through texting and empowered with AI. Another proposed virtual assistant reported "MoSHcA" that has developed the ability to make decisions on clinical considerations about a patient as per the input with or without human interventions.



10.4. Smart Hospital

Smart hospitals ease the processes from the patient as well as doctor perspectives. Remote patient monitoring allows patient-doctor interactions that limit the need of hospitals whereas device monitoring is linked with IoT improving the efficiency of the device and lowers the problem by early monitoring and hence reduces rescheduling and botheration of doctors. Nevertheless, the intravenous drug infusion and the rate of delivery has been reported to be monitored through narrowband IoT (NB-IoT) successfully in smart hospitals. Another IoT-based monitoring under Incentive Care Unit (ICU) has been reported by using INTEL GALILEO second-generation development board that not only enhances the communication between the patient and the medical professions but also record blood pressure, heart rate, temperature, and ECG. Under emergency conditions, the system sends the generated to the patient's doctor with medical conditions of the patient.

10.5. Assist in Research and Development

For drug discovery and ongoing research for various diseases, health monitoring devices play a vital role. Patient's bioanalysis can be analyzed through wearable devices. These devices monitor a number of essential biometrics inclusive of step count, calorie burnt, heart rate, and sleep, and specific wearables also detect glucose levels, sweat analysis, pressure sensors, and even some the patches have also been reported that can be used for cardiovascular disease, respiratory and stress disorders. The outcome pattern of the specific patients helps in clinical trials as well as reduce the cost of drug development. Commonly used wearable includes Empatica used for Epilepsy monitoring, Fitbit, epidermal iontophoretic biosensors sweat pattern detections, and many more.

11. Challenges and some improvement ideas

While the use of IoT (Internet of Things) in Medicine brings many benefits, there are also some challenges that need to be addressed. The accuracy and reliability of data collected by IoT devices are crucial for effective decision-making in healthcare. However, there may be situations where IoT devices produce inaccurate or inconsistent data due to device limitations, calibration issues, or user error. Adequate calibration, regular maintenance, and validation processes should be implemented to ensure the accuracy and reliability of IoT-generated data.

Privacy is provided by standardized data security protocols (such as encryption, authentication, and key distribution) during the data collection phase. In cases where security protocols are not suitable for IoT devices or cannot meet the application requirements, specific encryption, authentication, or key distribution algorithms are proposed. In these situations, there may also be issues related to indirect privacy leaks, such as finding user behavior patterns and protecting user anonymity. In the data transmission and storage/sharing phases, the individual system design must be made according to specific policies or rules that provide privacy to system users.

IoT devices and associated applications should be user-friendly and intuitive to ensure compliance and patient engagement. Complex user interfaces or difficult-use devices may deter patients from actively participating in their health programs. Designing devices and interfaces with the end user

in mind, conducting usability tests, and collecting user feedback can help improve the overall user experience.

IoT devices collect and transmit sensitive patient data, including personal health information. Ensuring the security and confidentiality of this data is crucial to protecting patient privacy. Healthcare providers must implement robust security measures such as encryption, secure data storage, access control, and regular system updates to protect patient data from unauthorized access or cyber threats. The solution is to develop uniform data quality standards. In this way, it can be managed more efficiently across countries, organizations, and departments, thereby facilitating the storage, delivery, and sharing of data and reducing errors in judgment and decision-making due to data incompatibility, data redundancy, and data deficiencies. Since IoT systems are distributed in nature, the use of international standards can have a positive effect on improving the performance of business processes by aligning different organizations with the same foundation, addressing interoperability issues, and ultimately working in a seamless manner.

IoT devices in healthcare collect extensive data about patient's movements, health conditions, and activities. Healthcare providers must address ethical concerns about data ownership, consent, and use. Patients must be informed about the data collected, how it will be used, and have control over their data. Implementing clear data governance policies and complying with relevant privacy regulations are essential. Healthcare providers need to assess the cost implications and ensure the availability of necessary infrastructure and resources to support IoT implementation effectively. Addressing these challenges requires collaboration among healthcare providers, technology developers, regulatory bodies, and other stakeholders. By addressing security concerns, promoting interoperability, ensuring data accuracy, and prioritizing user experience, IoT can realize its potential to transform medical sector while providing safe and effective patient care. (Fuior et al., 2023)

12. Conclusion

This review provides a comprehensive overview of the Internet of Things (IoT) in the context of healthcare, exploring its definition, architecture, and diverse applications. The IoT is defined as a global infrastructure facilitating advanced services through the interconnectedness of physical and virtual elements. The expansive definition encompasses a wide range of objects, from appliances to sensors, linked through evolving information and communication technologies.

The article delves into the application areas of the IoT, emphasizing its transformative impact on various sectors. Healthcare emerges as a significant beneficiary, with IoT-enabled devices revolutionizing patient monitoring, hospital operations, disease diagnosis, and treatment. The review explores IoT applications in diverse healthcare aspects, such as remote patient monitoring, management of hospital operations, glucose monitoring, depression detection, cancer detection, Parkinson's monitoring, ingestible sensors, robotic surgeries, and more.

Furthermore, the integration of IoT with wearables is explored, showcasing how smart healthcare devices, such as smartwatches and bracelets, contribute to disease detection and monitoring. The discussion extends to the broader smart healthcare system, encompassing AI, cloud computing, and nanoelectronics, with a focus on disease diagnosis, health risk monitoring, voice assistants, smart hospitals, and support for research and development.



Despite the significant benefits, the review acknowledges and addresses challenges associated with IoT implementation in healthcare. These challenges include data accuracy and reliability, privacy concerns, user-friendliness, security, ethical considerations, and the need for standardized data quality standards. Collaboration among stakeholders, including healthcare providers, technology developers, and regulatory bodies, is emphasized as crucial for overcoming these challenges.

In summary, while the IoT in healthcare presents immense potential for improving patient care, challenges must be systematically addressed to ensure the secure, effective, and ethical implementation of IoT technologies in the medical sector.

13. References

- Al-Fuqaha, A., Guizani, M., Mohammadi, M., Aledhari, M., & Ayyash, M. (2015). Internet of Things: A Survey on Enabling Technologies, Protocols, and Applications. *IEEE Communications Surveys & Tutorials*, 17(4), 2347-2376.
<https://doi.org/10.1109/COMST.2015.2444095>
- Atlam, H. F., Walters, R. J., & Wills, G. B. (2018). Internet of Things: State-of-the-art, Challenges, Applications, and Open Issues. *International Journal of Intelligent Computing Research*, 9(3), 928-938.
<https://doi.org/10.20533/ijicr.2042.4655.2018.0112>
- Fuor, R., Sălceanu, A., Luca, C., & Corciovă, C. (2023). Application of Internet of Things (IoT) in Biomedicine: Challenges and Future Directions. *En Internet of Things—New Insights [Working Title]*. IntechOpen.
<https://doi.org/10.5772/intechopen.113178>
- Mangla, M., Sharma, N., Mittal, P., Wadhwa, V. M., Kannapiran, T., & Khan, S. (Eds.). (2021). *Emerging technologies for healthcare: Internet of things and deep learning models*. Scrivener Publishing.
- Verma, D., Singh, K. R., Yadav, A. K., Nayak, V., Singh, J., Solanki, P. R., & Singh, R. P. (2022). Internet of things (IoT) in nano-integrated wearable biosensor devices for healthcare applications. *Biosensors and Bioelectronics: X*, 11, 100153.
<https://doi.org/10.1016/j.biosx.2022.100153>
- Víctor Manuel Brea Sánchez. (2018). *Internet_de_las_Cosas_Horizonte_2050.pdf*. <https://www.ieee.es/publicaciones-new/documentos-de-investigacion/2018/DIEEINV17-2018.html>

